

Calculus of variations: motivations towards classical field theory

Presentation - CUMC 2026

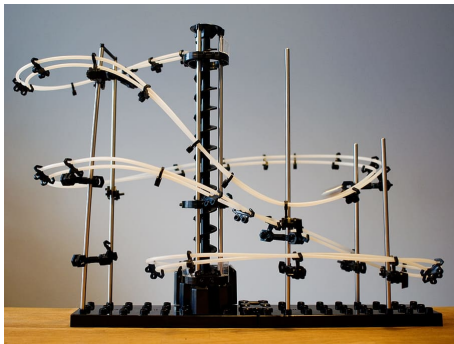
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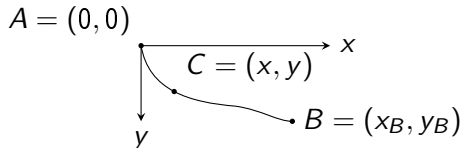
June 27, 2026

The brachistochrone curve problem

- Problem: Given two points at different heights, find the path of fastest descent under gravity, without friction (the **brachistochrone curve**).
- Galileo, 1638: conjectures a circular arc
- Johann Bernoulli, 1696: poses the problem clearly
- I. Newton, G. Leibniz and others: find independent solutions



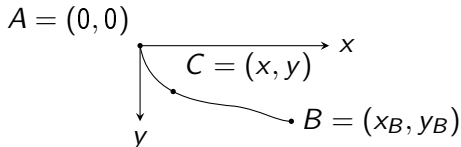
Mathematical formulation



- The curve joining A to B is defined by a function

$$y : [0, x_B] \rightarrow \mathbb{R}$$

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- We place a ball of mass m at A , at rest.
- Applying the conservation of energy between A and $C = (x, y(x))$, with $E_C = \frac{1}{2}mv_C^2 - mgy_C = \frac{1}{2}mv(x)^2 - mgy(x)$, we get:

$$\frac{1}{2}mv(x)^2 - mgy(x) = 0 \implies v(x) = \sqrt{2gy(x)}$$

Mathematical formulation

We would like to minimise the total time travelled, given by

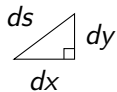
$$T[y] = \int_0^{t_B} dt.$$

Mathematical formulation

We would like to minimise the total time travelled, given by

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where the infinitesimal length is given by the Pythagorean theorem



so that

$$ds = \sqrt{dx^2 + dy^2} = \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx = \sqrt{1 + y'^2} dx$$

and

$$dt = \frac{ds}{v} = \sqrt{\frac{1 + y'^2}{2gy}} dx$$

Mathematical formulation

Therefore, the function to minimise is defined by

$$\begin{aligned} T : \mathcal{A} &\longrightarrow (0, +\infty) \\ y &\longmapsto \int_0^{x_B} L(y(x), y'(x)) dx \end{aligned}$$

with

$$\mathcal{A} = \{y \in \mathcal{C}^1([0, x_B], [0, +\infty)) : y(0) = 0, y(x_B) = y_B\}$$

and

$$L(y, y') = \sqrt{\frac{1 + y'^2}{y}} \quad \equiv \quad L(q, v) = \sqrt{\frac{1 + v^2}{q}}$$

(the constant $\frac{1}{\sqrt{2g}}$ is not relevant for the optimisation problem).

Calculus of variations: basic definitions

For $f : \mathbb{R} \rightarrow \mathbb{R}$, recall that if we can write $f(x + \varepsilon) = f(x) + \varepsilon g(x) + O(\varepsilon^2)$, then $g(x) = \lim_{\varepsilon \rightarrow 0} \frac{f(x + \varepsilon) - f(x)}{\varepsilon} =: f'(x)$.

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Definition

- A **functional** is a function $S : \mathcal{B} \rightarrow \mathbb{R}$ where $\mathcal{B}$ is a space of functions. Typically, $\mathcal{B}$ is a normed vector space, or, more generally, an affine space modeled on a normed vector space.
- For $S : \mathcal{B} \rightarrow \mathbb{R}$ a functional, the idea is similar. For $h \in V_{\mathcal{B}}$ and $\varepsilon \in \mathbb{R}$, if we can write

$$S[y + \varepsilon h] = S[y] + \varepsilon R[y; h] + O(\varepsilon^2),$$

then S is **differentiable**, with **variation** (or Gâteaux derivative)

$$\delta S[y; h] = R[y; h] = \lim_{\varepsilon \rightarrow 0} \frac{S[y + \varepsilon h] - S[y]}{\varepsilon}.$$

Optimisation

Definition

A differentiable functional $S : \mathcal{B} \rightarrow \mathbb{R}$ has an **extremum** at $\hat{y} \in \mathcal{B}$ if there exists $\varepsilon > 0$ such that :

$$\forall y \in \mathcal{B}, \|y - \hat{y}\| < \varepsilon, S[y] \geq S[\hat{y}] \quad (\text{minimum})$$

or

$$\forall y \in \mathcal{B}, \|y - \hat{y}\| < \varepsilon, S[y] \leq S[\hat{y}] \quad (\text{maximum})$$

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Theorem (Critical point theorem)

If a differentiable functional $S : \mathcal{B} \rightarrow \mathbb{R}$ has an extremum at $\hat{y} \in \mathcal{B}$, then:

$$\delta S[\hat{y}; h] = 0$$

for all admissible h .

Proof of the critical point theorem

Suppose that S has a minimum at $\hat{y} \in \mathcal{B}$. Then, there exists $r > 0$ such that for all $y \in \mathcal{B}$ with $\|y - \hat{y}\| < r$, we have $S[y] \geq S[\hat{y}]$.

Take an admissible $h \in \mathcal{B}$. For $\varepsilon > 0$ such that $\varepsilon < \frac{r}{\|h\|+1}$, we get that

$$S[\hat{y} + \varepsilon h] \geq S[\hat{y}] \quad , \quad S[\hat{y} - \varepsilon h] \geq S[\hat{y}].$$

Thus, since S is differentiable, we obtain that:

$$\delta S[\hat{y}; h] = \lim_{\varepsilon \rightarrow 0^+} \frac{S[\hat{y} + \varepsilon h] - S[\hat{y}]}{\varepsilon} \geq 0$$

and

$$\delta S[\hat{y}; h] = \lim_{\varepsilon \rightarrow 0^-} \frac{S[\hat{y} + \varepsilon h] - S[\hat{y}]}{\varepsilon} = \lim_{\varepsilon \rightarrow 0^+} \frac{S[\hat{y} - \varepsilon h] - S[\hat{y}]}{-\varepsilon} \leq 0.$$

It follows that $\delta S[y; h] = 0$.

If $\hat{y}$ is a maximum, the proof is identical. $\square$

Some functionals have more applications than others. Here, we will be interested in functionals that take the form

$$S[y] = \int_a^b L(x, y, y') dx$$

called an **action**, and L a **Lagrangian** (y and y' are evaluated at x). Moreover, we choose the domain of S as

$$\mathcal{A}_{A,B} = \{y \in \mathcal{C}^1([a, b], \mathbb{R}) : y(a) = A, y(b) = B\}.$$

and take $L : \mathbb{R}^3 \rightarrow \mathbb{R}$ to be $\mathcal{C}^2$. We would like to compute the variation of S .

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Fixing $\varepsilon > 0$ and $h \in \mathcal{A}_{0,0}$ (the space of translations of the affine space $\mathcal{A}_{A,B}$), note that:

$$S[y + \varepsilon h] - S[y] = \int_a^b (L(x, y + \varepsilon h, y' + \varepsilon h') - L(x, y, y')) dx.$$

The integrand can be replaced by its first-order Taylor expansion.

$$S[y + \varepsilon h] - S[y] = \int_a^b (L(x, y + \varepsilon h, y' + \varepsilon h') - L(x, y, y')) dx$$

$$\begin{aligned}
 S[y + \varepsilon h] - S[y] &= \int_a^b (L(x, y + \varepsilon h, y' + \varepsilon h') - L(x, y, y')) \, dx \\
 &= \int_a^b \left(\nabla L(x, y, y')^T (0, \varepsilon h, \varepsilon h') + O(\|(0, \varepsilon h, \varepsilon h')\|^2) \right) \, dx
 \end{aligned}$$

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so that:

$$\begin{aligned}
 \delta S[y; h] &= \lim_{\varepsilon \rightarrow 0} \frac{S[y + \varepsilon h] - S[y]}{\varepsilon} \\
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 &= \int_a^b \nabla L(x, y, y')^T (0, h, h') dx \\
 &= \int_a^b \left(\frac{\partial L}{\partial y}(x, y, y') h + \frac{\partial L}{\partial y'}(x, y, y') h' \right) dx \\
 &= \int_a^b \left(\frac{\partial L}{\partial y}(x, y, y') - \frac{d}{dx} \frac{\partial L}{\partial y'}(x, y, y') \right) h dx + \left[\frac{\partial L}{\partial y'} h \right]_a^b.
 \end{aligned}$$

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Since we chose $h \in \mathcal{A}_{0,0}$, the second term vanishes. Moreover, if y is a critical point of S , the critical point theorem implies that

$$\int_a^b \left(\frac{\partial L}{\partial y}(x, y, y') - \frac{d}{dx} \frac{\partial L}{\partial y'}(x, y, y') \right) h dx = 0$$

for all $h \in \mathcal{A}_{0,0}$.

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for all $h \in \mathcal{A}_{0,0}$. Thinking it as a scalar product, we deduce that necessarily:

$$\frac{\partial L}{\partial y}(x, y, y') - \frac{d}{dx} \frac{\partial L}{\partial y'}(x, y, y') = 0.$$

To see this rigorously, one can choose the function

$$h(x) = [(x - x_1)(x - x_2)]^2 \mathbb{1}_{[x_1, x_2]}$$

where the function in parentheses is assumed by contradiction to have constant sign on $[x_1, x_2]$. Thus, the integral becomes nonzero.

The Euler-Lagrange equation

We have shown the following theorem

Theorem (Euler-Lagrange)

Let S be a functional

$$S[y] = \int_a^b L(x, y, y') dx$$

defined for $y \in \mathcal{A} = \{y \in C^1([a, b], \mathbb{R}) : y(a) = A, y(b) = B\}$ and L being a C^2 function.

If y is a critical point of S , then y is a solution of the **Euler-Lagrange equation**

$$\frac{\partial L}{\partial y} - \frac{d}{dx} \frac{\partial L}{\partial y'} = 0 \quad \equiv \quad \frac{\partial L}{\partial q} - \frac{d}{dx} \frac{\partial L}{\partial v} = 0 \quad (\text{EL})$$

Application: the brachistochrone

Recall that solving the brachistochrone problem is mathematically equivalent to minimising the action

$$T[y] = \int_0^{x_B} L(x, y, y') dx \quad , \quad L(x, y, y') = \sqrt{\frac{1 + y'^2}{y}}.$$

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First, the application of the chain rule to L

$$\frac{d}{dx} \left(L - y' \frac{\partial L}{\partial y'} \right) = \frac{\partial L}{\partial x} + \frac{\partial L}{\partial y} y' + \frac{\partial L}{\partial y'} y'' - y'' \frac{\partial L}{\partial y'} - y' \frac{d}{dx} \frac{\partial L}{\partial y'}$$

rewrites (EL) as:

$$\frac{d}{dx} \left(L - y' \frac{\partial L}{\partial y'} \right) = \frac{\partial L}{\partial x}$$

Note that the RHS vanishes since L does not depend explicitly on x . Hence, introducing a constant of integration C , (EL) reduces to

$$L - y' \frac{\partial L}{\partial y'} = C$$

Computation of the solution

Starting from $L(x, y, y') = \sqrt{\frac{1+y'^2}{y}}$, we obtain that

$$\frac{\partial L}{\partial y'} = \frac{y'}{\sqrt{y(1+y'^2)}}.$$

Plugging this into the modified (EL), $L - y' \frac{\partial L}{\partial y'} = C$, we obtain

$$\sqrt{\frac{1+y'^2}{y}} - \frac{y'}{\sqrt{y(1+y'^2)}} = C$$

and reduces to

$$y(1+y'^2) = \frac{1}{C^2} := K \implies y' = \sqrt{\frac{y}{K-y}}$$

which is a separable ODE.

The final solution

- The final solution can be expressed in a parametric form. In fact, considering the initial position of the ball, one obtains

$$\begin{cases} x(\theta) &= k(\theta - \sin \theta) \\ y(\theta) &= k(1 - \cos \theta) \end{cases}$$

with $k := K/2$ chosen such that $(x(\theta), y(\theta))$ passes through the point $B = (x_B, y_B)$.

- The curve obtained is called a **cycloid**.
- Animation, cycloid
- Of course, T does not have a maximum, hence this critical point is a minimum.
- Animation, brachistochrone

Field theory: the covariant phase space

Definition

A **classical field theory** consists of:

- An n -dimensional **manifold** M (the space-time)
- A set of **fields** $\mathcal{F}$ (the set of smooth sections of a fiber bundle $E \xrightarrow{\pi} M$).
For example, scalar fields, tensor fields, spinor fields.
- A **Lagrangian** $L : \mathcal{F} \rightarrow C^\infty(M)$ that defines the **action** $\mathbb{S}(L) = \int_M L d^n x$.

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-
- It is possible to define the variation δ of any Lagrangian, and think of it as the differential operator of the exterior derivative on the infinite-dimensional manifold $\mathcal{F}$.
 - Thus, one defines the **space of solutions** with the same idea as before:

$$\text{Sol}(\mathbb{S}) := \{\phi \in \mathcal{F} : \delta_\phi \mathbb{S} = 0\}$$

and this space describes the equations of motion.

Example 1: Scalar field

- (M, g) is an arbitrary manifold with a fixed metric g .
- $\mathcal{F} = C^\infty(M)$ is the set of smooth $\mathbb{R}$ -valued functions defined on M .
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$$L(\phi) = \frac{1}{2}g^{-1}(d\phi, d\phi) = \frac{1}{2}g^{\mu\nu}\partial_\mu\phi\partial_\nu\phi = \frac{1}{2}\partial_\mu\phi\partial^\mu\phi$$

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One can compute the equations of motion with a formula similar to (EL), or simply vary the action and use Stokes' theorem. We obtain that:

$$0 = \delta_\phi \mathbb{S} = \int_M g^{\mu\nu} \partial_\mu(\delta\phi) \partial_\nu\phi \, d^n x = \dots$$

After some computations, one obtains:

$$\operatorname{div}(\vec{\nabla}\phi) = 0 \quad \equiv \quad \square\phi = 0$$

which is the **wave equation**! For instance, with the Minkowski metric $g_{\mu\nu} = \operatorname{diag}(-1, 1, 1, 1)$, it follows that:

$$-\frac{\partial^2\phi}{\partial t^2} + \frac{\partial^2\phi}{\partial x^2} + \frac{\partial^2\phi}{\partial y^2} + \frac{\partial^2\phi}{\partial z^2} = 0$$

Example 2: General Relativity (Einstein-Hilbert bulk lagrangian)

- M is an n -dimensional manifold without boundary
- $\mathcal{F} = \text{LMet}(M)$ is the set of Lorentzian metrics on M (a subset of the $(0, 2)$ tensor fields).
- The (Einstein-Hilbert) Lagrangian is given by $L_{EH}(g) = R_g \sqrt{-\det(g)}$ (where R_g is the scalar curvature of (M, g)), so that:

$$\mathbb{S}[g] = \int_M R_g \sqrt{-\det(g)} \, d^4x.$$

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Computing $\delta_g \mathbb{S}$ and setting it equal to 0, one obtains the (vacuum) **Einstein field equations**:

$$R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu} = 0$$

where $R_{\mu\nu}$, R provides information on the curvature of the spacetime (M, g) .

Adding boundaries

- In the proof of (EL), remember that, we cancelled the term

$$\left[\frac{\partial L}{\partial y'} h \right]_a^b.$$

- On the manifold M , the "integration by parts" leads a the term of the form:

$$\Omega = \int_{\partial \Sigma} \iota^* \delta \Theta$$

that provides informations (e.g. conserved quantities) on the boundary (a presymplectic form on $\mathcal{F}$), where $\iota : \Sigma \hookrightarrow M$. Ω is totally canonical if pulled-back in $\text{Sol}(\mathbb{S})$.

- Showing that Ω is independent of the choice of Θ is not straightforward in presence of boundaries (because of the Stokes' theorem).
- Studying boundaries has various applications in physics (e.g. black holes).

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